

A quantitative-aperture reduction from nontangential RMF to dyadic RMF

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Status. Conditional reduction, likely valid, for expert review. The argument below proves the stated implication from an explicit quantitative aperture hypothesis. It does *not* prove that ordinary nontangential RMF supplies that hypothesis, and it does not prove the reverse implication.

1 Source question

The source is F. Di Plinio and Y. Ou, *Banach-valued multilinear singular integrals*, arXiv:1506.05827v3, Remark 3.7 on printed page 14 [1]. The authors ask whether dyadic RMF and nontangential RMF are equivalent.

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In the same article [15], the authors prove that the dyadic RMF property holds for the same classes of Banach spaces appearing in Lemma 3.6. At present time, we are unaware of further classes of Banach spaces for which the dyadic RMF property, as well as the nontangential RMF property holds true. It seems natural to conjecture that dyadic RMF and nontangential RMF are indeed equivalent. This is mentioned in Kempainen's thesis [27] as an open question. While we are unable to derive either implication of this equivalence, the above results entail that dyadic RMF and nontangential RMF are now known to hold in exactly the same generality.

4. THE COIFMAN-MEYER THEOREM IN UMD SPACES

In this section, we present our main operator-valued multiplier theorem, Theorem 1 below.

Fix the triple of Banach spaces and the trilinear identification used in the source. For a family $S \subset X_3$, write $\mathcal{R}(S)$ for the corresponding Rademacher bound. Let Φ be the normalized grand class of kernels from (3.1) of the source, and put

$$\mathcal{M}_{\Phi, \alpha} f(x) = \sup_{\phi \in \Phi} \mathcal{R}\{f * \phi_t(y) : |y - x| < \alpha t, t > 0\}.$$

For the canonical dyadic lattice $\mathcal{D}$, put

$$\mathcal{M}^{\Delta} f(x) = \mathcal{R}\{\langle f \rangle_Q : Q \in \mathcal{D}, x \in Q\}.$$

Multiplying the defining seminorm of Φ by a fixed constant changes only harmless constants below.

2 Idea of the proof

A sharp cube average can be reached from a smooth cube average by letting the smoothing scale tend to zero. Differentiating in that scale and integrating by parts moves the derivative to the boundary of the cube. At relative scale 2^{-m} , the boundary term has total mass $O(2^{-m})$, while all of its smooth averages lie in a cone of aperture $O(2^m)$ above every point of the cube. Thus

$$\mathcal{M}^\Delta f \lesssim \mathcal{M}_{\Phi, C} + \sum_{m \geq 0} 2^{-m} \mathcal{M}_{\Phi, C 2^m} f.$$

The displayed pointwise estimate is unconditional. The conditional dependency enters only when taking L^p norms and summing the aperture constants.

3 The reduction

For $1 < p < \infty$ and $\alpha \geq 1$, denote the aperture norm by

$$A_p(\alpha) = \|\mathcal{M}_{\Phi, \alpha}\|_{L^p(\mathbb{R}^d; X_3) \rightarrow L^p(\mathbb{R}^d)}.$$

Theorem 1 (summable-aperture criterion). *There are dimensional constants $c_d, C_d > 0$ such that, for every Schwartz function $f : \mathbb{R}^d \rightarrow X_3$,*

$$\mathcal{M}^\Delta f(x) \leq C_d \left(\mathcal{M}_{\Phi, c_d} f(x) + \sum_{m=0}^{\infty} 2^{-m} \mathcal{M}_{\Phi, c_d 2^m} f(x) \right) \quad (1)$$

for almost every x . Consequently, if

$$A_p(c_d) + \sum_{m=0}^{\infty} 2^{-m} A_p(c_d 2^m) < \infty, \quad (2)$$

then X_3 has dyadic RMF_p . In particular, it is enough that

$$A_p(\alpha) \leq C \alpha^\theta \quad (\alpha \geq 1)$$

for some $\theta < 1$.

Proof. Choose $\eta \in C_c^\infty(\mathbb{R}^d)$ with $\int \eta = 1$. After one fixed normalization, η and

$$\phi_i(u) = u_i \eta(u), \quad i = 1, \dots, d,$$

belong to the grand kernel class. If Q is a cube of side length ℓ and $r > 0$, define

$$H_Q(r) = \frac{1}{|Q|} \int_Q (f * \eta_r)(y) dy.$$

For Schwartz f , $H_Q(r) \rightarrow \langle f \rangle_Q$ in X_3 as $r \downarrow 0$. Moreover,

$$\partial_r \eta_r = - \sum_{i=1}^d \partial_i (\phi_i)_r.$$

The divergence theorem, valid here for the smooth X_3 -valued convolutions, therefore gives

$$H'_Q(r) = -\frac{1}{|Q|} \sum_{i=1}^d \int_{\partial Q} (f * (\phi_i)_r)(y) n_i(y) d\sigma(y). \quad (3)$$

It follows that

$$\langle f \rangle_Q = H_Q(\ell) + \sum_{m=0}^{\infty} B_{m,Q}, \quad B_{m,Q} = - \int_{2^{-m-1}\ell}^{2^{-m}\ell} H'_Q(r) dr. \quad (4)$$

We use the standard facts that Rademacher bounds are unchanged by passing to the closed absolutely convex hull, are preserved by normalized Bochner averaging, and are subadditive under Minkowski sums; these are also used in the source (notably in its proof of Lemma 3.2).

Fix x and a finite collection $Q_1, \dots, Q_N \in \mathcal{D}$ containing x . Since $|y - x| \leq \sqrt{d}\ell(Q_j)$ for $y \in Q_j$, normalized averaging in the definition of $H_{Q_j}(\ell(Q_j))$ yields

$$\mathcal{R}\{H_{Q_j}(\ell(Q_j)) : 1 \leq j \leq N\} \leq C_d \mathcal{M}_{\Phi, c_d} f(x). \quad (5)$$

For $r \in (2^{-m-1}\ell(Q_j), 2^{-m}\ell(Q_j)]$ and $y \in \partial Q_j$, one has

$$|y - x| \leq \sqrt{d}\ell(Q_j) \leq 2^{m+1}\sqrt{d}r.$$

On the other hand, the total variation of the normalized boundary-scale measure in (3), restricted to this interval of r , is at most

$$\frac{|\partial Q_j|}{|Q_j|}, 2^{-m-1}\ell(Q_j) = d, 2^{-m}.$$

Splitting the d kernel components and using preservation under Bochner averaging therefore gives

$$\mathcal{R}\{B_{m,Q_j} : 1 \leq j \leq N\} \leq C_d 2^{-m} \mathcal{M}_{\Phi, c_d 2^m} f(x). \quad (6)$$

Combining (4)–(6), using subadditivity, and then letting the finite collection exhaust the dyadic cubes through x proves (1). Minkowski's inequality and (2) give

$$\|\mathcal{M}^\Delta f\|_p \leq C_d \left(A_p(c_d) + \sum_{m \geq 0} 2^{-m} A_p(c_d 2^m) \right) \|f\|_{L^p(X_3)}.$$

Density completes the L^p statement. If $A_p(\alpha) \leq C\alpha^\theta$ with $\theta < 1$, the remaining series is geometric. $\square$

4 Conditional dependencies and limitations

The boundary reconstruction and the pointwise estimate (1) are complete. The unproved dependency is precisely (2): ordinary nontangential RMF gives finiteness of each fixed-aperture norm but no summable quantitative growth estimate is currently derived here. The condition cannot simply be replaced by aperture-uniform boundedness without justification. Already in the scalar case, for a nonnegative compactly supported approximate identity and a test function supported on a unit ball, the aperture- α maximal function stays bounded below on a region of volume comparable to α^d ; hence its L^p norm has growth at least of order $\alpha^{d/p}$.

Even after proving (2), this packet establishes only the nontangential-to-dyadic implication. The dyadic-to-nontangential implication remains untouched.

The most important reviewer task is to check the passage from the boundary formula (3) to the simultaneous Rademacher-bound estimate (6), including the normalization of the finitely many kernels $u_i \eta(u)$ into the source's grand class.

5 Verification and novelty status

The proof was checked symbolically: differentiating $r^{-d}\eta(z/r)$ gives $-r^{-d-1}(d\eta + (z/r) \cdot \nabla\eta)$, exactly $-\sum_i \partial_i(\phi_i)_r$. The boundary mass on the m th dyadic scale is $d2^{-m}$, and the required aperture is at most $2^{m+1}\sqrt{d}$. No numerical or finite-dimensional experiment is used.

A bounded search on 21 July 2026 covered the exact phrases “nontangential Rademacher maximal function”, “nontangential RMF”, and “dyadic RMF and nontangential RMF”, together with arXiv:1506.05827 and the later arXiv papers 1905.02139 and 1908.07233. No later arXiv paper explicitly settling the equivalence or stating this boundary/aperture reduction was found. This is not a proof of novelty.

References

- [1] F. Di Plinio and Y. Ou, *Banach-valued multilinear singular integrals*, Indiana Univ. Math. J. 67 (2018), no. 5, 1711–1763; arXiv:1506.05827.